Solid-State Structures

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Fibrous Red Phosphorus**

Michael Ruck, Diana Hoppe, Bernhard Wahl, Paul Simon, Yuekui Wang, and Gotthard Seifert*

According to general chemistry textbooks, three basic allotropic modifications of elemental phosphorus exist, namely, white, red, and black phosphorus, which differ considerably in their chemical and physical properties. When white phosphorus is heated or exposed to radiation, it is transformed to amorphous red phosphorus. The structure of red phosphorus is considered to be a polymeric network of different building units, which are well-known from various compounds containing polymeric phosphorus structures.[1,2] While this network features only short or intermediate range order, further heating or irradiation leads to crystallization. Different crystalline allotropes were reported to form in this way, for example, red phosphorus IV and red phosphorus V, which were characterized by X-ray powder diffraction.[3]

Already in 1865 Hittorf found that phosphorus can also be crystallized from a melt of lead, giving red-to-violet colored platelets.[4] In 1969 Thurn and Krebs determined the structure of this crystalline type of phosphorus, called violet or Hittorf's phosphorus.[5] Up to now, it is the only structurally characterized crystalline form of red phosphorus. This modification consists of pentagonal tubes formed by covalently bonded phosphorus atoms. By comparison of the powder diffraction patterns it can be equated with red phosphorus V described earlier.

Furthermore, Thurn and Krebs reported in their work on the structure of Hittorf's phosphorus: "When white phosphorus is annealed at temperatures between 545 and 580 °C for several months, in addition to platelike crystals of Hittorf's phosphorus bunches of capillary parallel lying fibers form. […]".[5] From X-ray rotation photographs of the fibers, they proposed an arrangement of tubes of the same kind as in Hittorf's phosphorus but linked to parallel double tubes. Furthermore, they considered the constitution to be energetically stable but estimated the density to be lower than the density of Hittorf's phosphorus. Later Böcker and Häser showed in their theoretical investigations on phosphorus structures that Hittorf's phosphorus and the proposed structure of the fibrous type are energetically equivalent and the most favored of all the possible phosphorus chains.[6]

We have now confirmed the assumed structure of the fibrous type of red phosphorus using single-crystal X-ray diffraction as well as transmission electron microscopy methods. The iodine-catalyzed recrystallization of red phosphorus in a small temperature gradient yields dark red needles. In most cases the crystal quality of the fibrous phosphorus is very poor due to agglomeration and stacking faults. Frequently radial growth of fiber bunches upon the ampule wall was observed. Furthermore, needles and platelets often share edges, indicating an intergrowth of crystals of Hittorf's phosphorus and of the fibrous type. The two types can hardly be distinguished visibly; both show a bright red to deep red color and a more or less platelike shape. However, under mechanical stress crystals of the fibrous type split into thin filaments, whereas the crystal platelets of the Hittorf phosphorus type just break up into smaller platelets. In addition, the crystallization of red phosphorus from a lead melt[4,5] was carried out in order to examine whether crystals of the fibrous type can be obtained. But exclusively crystals of the Hittorf type formed by this means.

Powder diffraction patterns of sublimation products show reflections that can be assigned to Hittorf's phosphorus and numerous additional reflections. These can be assigned entirely to the fibrous type after comparison to the theoretical pattern calculated based on structural data. On comparison of the powder diffraction pattern of fibrous phosphorus with powder diffraction patterns given in ref. [3], it becomes clear that the so-called form IV of red phosphorus corresponds to the fibrous type presented here.

Under the transmission electron microscope the two crystalline forms can be differentiated easily: In Figure 1 a the high-resolution micrograph of perpendicular strands is indicative of Hittorf's phosphorus; Figure 1 b in contrast shows exclusively strands lying parallel to each other, which

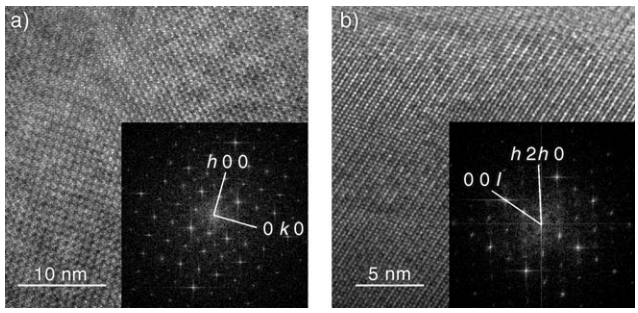

Figure 1. a) High-resolution electron micrograph showing the pseudo-tetragonal symmetry of the [001] zone of Hittorf's phosphorus at the Scherzer focus (Philips CM200 microscope), plane (*hk*0); the corresponding FFT is shown in the inset. b) High-resolution electron micrograph of fibrous phosphorus (Philips CM30 microscope), plane (*h*,2*h*,*l*); the corresponding FFT is shown in the inset.

[*] Prof. Dr. M. Ruck, D. Hoppe, B. Wahl
Institut für Anorganische Chemie
Technische Universität Dresden
01062 Dresden (Germany)
Fax: (+49) 351-463-37287
E-mail: michael.ruck@chemie.tu-dresden.de
Dr. P. Simon
Max-Planck-Institut für Chemische Physik fester Stoffe
Nöthnitzer Strasse 40, 01187 Dresden (Germany)
Dr. Y. Wang, Prof. Dr. G. Seifert
Institut für Physikalische Chemie und Elektrochemie
Technische Universität Dresden
01062 Dresden (Germany)

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corresponds to fibrous phosphorus. Fast Fourier transform (FFT) of the micrographs of Hittorf's phosphorus gives images with orthogonal rows of pseudo-Bragg reflections (plane $hk0$) and reciprocal lattice parameters of $a^* \approx b^* \approx (920\,\mathrm{pm})^{-1}$. The FFT images of fibrous phosphorus (plane $(h,2h,l)$) give lattice parameters of $|\boldsymbol{a}^* + 2\boldsymbol{b}^*| \approx (440\,\mathrm{pm})^{-1}$ and $c^* \approx (580\,\mathrm{pm})^{-1}$; the included angle amounts to 55.5°. The c^* axis is perpendicular to the visible fiber direction.

The crystal structure of the fibrous type of crystalline red phosphorus is closely related to the structure of Hittorf's phosphorus. Both consist of tubes with pentagonal cross section, which in turn are a regular sequence of three different building units (Figure 2). The subunits are cages of eight or nine phosphorus atoms, which we denote as *[P8]* and *[P9]* in keeping with ref. [6], and dumbbells of two phosphorus atoms, *P2*.

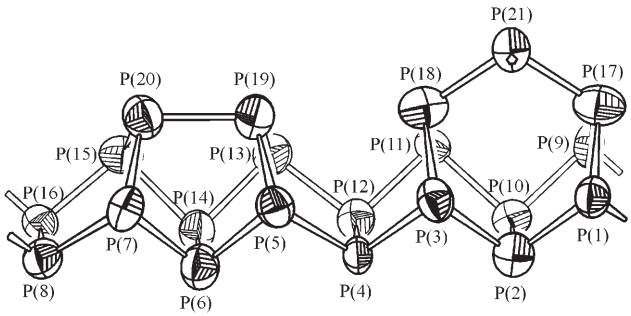

Figure 2. Repeating unit of the tubes in fibrous and in Hittorf's phosphorus (numbering of atoms in analogy to ref. [5]). Thermal ellipsoids enclose 90% probability for atomic displacement. Bond lengths are between 219(1) and 231(1) pm; the mean bond length is 221.9 pm.

The tubes can be interpreted alternatively as zigzag-shaped double chains, which form a corrugated ladder. Free valences are saturated by "roofs" of two or three phosphorus atoms. These single tubes feature the symmetry of the rod group $p2mm$ (no. 18).[7] At the medial roof atoms of the *[P9]* cages, denoted as P(21) and P(21^{i}) (Figure 3), two tubes are each connected to form double tubes with $p112/m$ symmetry (no. 11).[7] Covalent bonds of the atoms P(21) and P(21^{i}) are arranged in staggered conformation, and the corresponding lone pairs are aligned antiparallel. The arrangement resembles the energetically most favored conformation of methyl groups in 2,3-dimethylbutane.

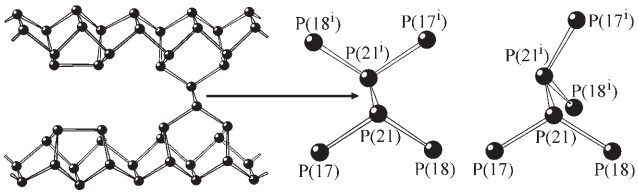

Figure 3. Linkage of the tubes in fibrous phosphorus (left; i: $-x$, $1-y$, $-z$) vs. Hittorf's phosphorus (right; i: $-x$, y, $0.5-z$). Bonding lengths and angles in the structures of fibrous and Hittorf's phosphorus[8] that differ by more than 3 SD: P(21)—P(21^{i}) 222.4(8)/217.3(1) pm; P(17)-P(21)-P(18) 111.9(3)/115.98(3)°, P(17)-P(21)-P(21^{i}) 94.7(3)/96.71(4)°, P(18)-P(21)-P(21^{i}) 93.6(3)/108.23(2)°.

In contrast, the connected tubes of Hittorf's phosphorus run perpendicularly, forming double layers of two interpenetrating but independent tube systems. These double layers are stacked in the c direction and held together by van der Waals forces. As a result of the perpendicular arrangement of two linked tubes, the described staggered conformation of the atom groups P(17)-P(21)-P(18) and P(17^{i})-P(21^{i})-P(18^{i}) cannot be attained (Figure 3, right). The consequence is a higher tension, expressed in expanded bonding angles at the atom P(21) in Hittorf's phosphorus.

The double tubes of the fibrous-type structure are stacked in the a direction with a displacement of $^1/_8\,b$ and in the c direction with a displacement $^1/_4\,b$. A distorted hexagonal pattern for the single tubes results (Figure 4). As a consequence of the packing, the $p112/m$ symmetry of the double tubes is not transferred to the crystal lattice, and the space group symmetry is $P\bar{1}$. The packing density of the tubes in Hittorf's and in fibrous phosphorus is very similar, although the arrangements of the tubes differs considerably. In contrast to the assumption in ref. [5], the calculated X-ray densities of fibrous phosphorus and Hittorf's phosphorus both amount to 2.37 g cm^{-3}. The shortest van der Waals distance in Hittorf's phosphorus is 306 pm, between atoms P(14) and P(14^{ii}) [ii: $-1-x$, $-1-y$, $-z$];[8] in fibrous phosphorus the shortest van der Waals distance is 315 pm, also between atoms P(14) and P(14^{ii}) [ii: $1-x$, $-y$, $-z$]. The observation of diffuse X-ray scattering in the diffraction pattern of many phosphorus fibers indicates alternative stacking sequences.

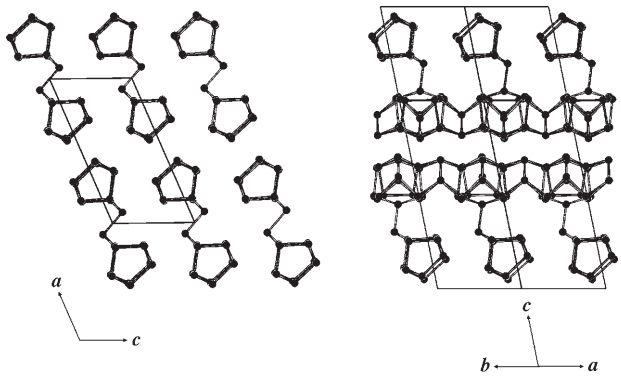

Figure 4. Left: Arrangement of the double tubes along the fiber axis in fibrous phosphorus. Right: Stacking of the layers in Hittorf's phosphorus.

Since the calculations of Böcker and Häser had focused on phosphorus clusters,[6] we have performed density functional based calculations for the bulk phases at the DFTB level[9] to investigate the relative stability of Hittorf's and fibrous phosphorus. The energy contributions of the van der Waals interactions were considered in a semiempirical a posteriori treatment.[10] Periodic boundary conditions were imposed on the systems in all calculations. Comparisons between the DFTB-optimized bond parameters and the experimental values can be found in the Supporting Informa-

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tion. Clearly, for both allotropes the calculated bond parameters are in quite good agreement with the experimental findings. For bond lengths, the standard deviations (SD) are about 1 %, and the maximum deviations (MD) are less than 2 %. The deviations in bond angles are SD = 0.8° and MD = 1.4° for Hittorf's phosphorus, and SD = 1.8° and MD = 3.6° for the fibrous structure. The results show that the energy differences for the two structures are all less than 0.003 eV per atom, regardless of which bond lengths (experimental or optimized) are used and whether the van der Waals interactions are included. Therefore no preferred structure could be figured out. The conclusion that neither structure is energetically preferred is in agreement with the findings of Böcker and Häser.[6]

For a better understanding of the bonding characteristics of fibrous and Hittorf's phosphorus, especially the short bridging bond P(21)–P(21^{i}) in the latter structure, we performed further calculations, using cluster models (Figure 5) at

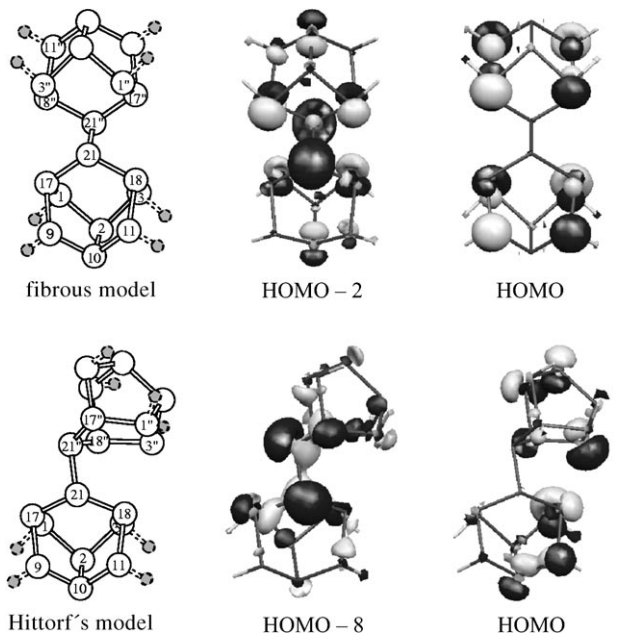

Figure 5. The cluster models ($P_{18}H_8$) used to simulate the bridge between the *[P9]* units in fibrous and in Hittorf's phosphorus (left). The highest occupied molecular orbitals (HOMO) and the orbitals close to the HOMO with dominant "lone-pair" character on the bridge atoms are shown on the right side. Note that in the configuration corresponding to Hittorf's phosphorus the orbital HOMO−8 exhibits some π-bonding characteristics between the bridge atoms, while in the fibrous structure no such contribution can be traced (see orbital HOMO−2).

the DFT/LSD/DZVP level. The fixed (X-ray) geometries of fibrous and Hittorf's phosphorus were used in these calculations. The free valences in these clusters were saturated with hydrogen (P−H = 142 pm). The results indicate that the orbitals around the HOMO show significant lone-pair character. In the fibrous structure one of these orbitals (HOMO−2), dominated by the lone-pair electrons located on the bridge atoms P(21), is only about 0.1 eV below the HOMO. In Hittorf's phosphorus the corresponding orbitals are at lower energy (1 eV below the HOMO). It seems that the twist structure in Hittorf's model is favorable to lower the energy of "lone-pair" electrons on the bridge atoms due to some π-bonding contribution, while in the fibrous structure the inversion symmetry prohibits such an effect. This accounts for the unusually short bridging bond in Hittorf's phosphorus and corresponds to the expanded bonding angles. In addition, a DFT geometry optimization of the Hittorf cluster model was also performed at the LSD/DZVP level, which confirmed that the bridging bond P(21)−P(21^{i}) is the shortest one (experimental 217.8 pm, calcd 219.5 pm).

As a general conclusion it can be stated that the fibrous type of crystalline red phosphorus bears strong structural resemblance to Hittorf's phosphorus. It represents a new crystalline modification of phosphorus. Both types of red phosphorus can be obtained in the same way. The growth mechanism is still unknown, yet experimental results indicate that homogeneous nucleation, for instance in a lead melt, leads only to Hittorf's phosphorus, whereas the growth of phosphorus fibers is preferred in case of heterogeneous nucleation. The structure model given by Thurn and Krebs as well as the theoretical predictions for the existence and stability of a fibrous type of phosphorus by Böcker and Häser thus could be confirmed.

Phosphorus tubes of similar architecture were identified in numerous compounds, such as polyphosphides (e.g. MP_{15}, M = Li–Cs;[11] $M'M''P_{14}$, M′ = Zn, Cd, Hg, M″ = Sn, Pb[12]); they represent quite stable possibilities of connecting phosphorus atoms.[6,11,13] Other compounds that contain exclusively polymeric tubular structures of phosphorus atoms embedded in copper halide matrices were scrutinized over the past decades.[14] Moreover, two of these phosphorus structures were extracted from their matrices.[15] The phosphorus architectures found in copper halide matrices differ from those obtained from pure amorphous red phosphorus, in all probability because the pentagonal tubes in Hittorf's and fibrous phosphorus are energetically more stable and can be arranged more densely in a crystal than the corrugated or helical phosphorus rods forming in the matrices.

Experimental Section

Synthesis: Amorphous red phosphorus (Fluka 99 %; purified according to ref. [16]) was sublimed in evacuated fused-silica ampules using iodine (Merck; sublimed twice on barium oxide) as a catalyst.[17] Best results were achieved with approximately 1.0 g of phosphorus and 40 mg of iodine in an ampule of 17 cm^3. Hittorf's phosphorus and a fibrous solid were obtained. The latter showed increasing crystallinity when the temperature was increased and the temperature gradient decreased within the ampule. Sufficiently large crystals could be grown in a temperature gradient of about 10 K in the temperature range between 590 and 570 °C over several days (see also ref. [18]). At higher temperatures the phosphorus vapor pressure increases rapidly due to the increasing formation of white phosphorus in the gas phase. Therefore the samples were not heated above the given temperatures. Sublimation of red phosphorus also succeeds without catalysts. However, the generation of a gas phase is kinetically hindered, and autosublimation requires much more time.

Transmission electron microscopy: Crystallites of fibrous red phosphorus and of Hittorf's phosphorus were examined by high-resolution transmission electron microscopy (HRTEM). The inves-

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tigations were performed at the Speziallabor für Elektronenmikroskopie Triebenberg with a Philips CM200 FEG\ST Lorentz electron microscope, equipped with a field emission gun and at an acceleration voltage of 200 kV. Point resolution amounts to 2.4 Å and information limit to about 1.2 Å. A TEM Philips CM30 FEG UT/Special Tübingen at an acceleration voltage of 300 kV also equipped with a field emission gun was also used. In this case, point resolution amounts to 1.7 Å and information limit to about 0.9 Å. TEM images were processed by Digital Micrograph (Gatan, USA).

X-ray crystal structure determination: systematic name: phosphorus ($aP42$); triclinic, space group $P\bar{1}$ (no. 2), $a = 1219.8(8)$, $b = 1298.6(8)$, $c = 707.5(7)$ pm; $\alpha = 116.99(7)$, $\beta = 106.31(7)$, $\gamma = 97.91(7)°$, $V = 911(2) \times 10^6$ pm^3, $\rho_{calcd} = 2.37$ g cm^{-3}, absorption coefficient $\mu = 1.89$ mm^{-1}, $2\theta_{max} = 42.5°$, $Mo_{K\alpha}$ radiation, $\lambda = 71.073$ pm, $T = 293(2)$ K. The diffraction data were collected on an imaging plate diffraction system IPDS-I (Stoe). Numerical absorption corrections were applied. The structure was solved with SHELXS97 and refined against F_o^2 using SHELXL97.[19] All atoms were refined using anisotropic displacement parameters. Final R values: $R_1(F_o > 4\sigma(F_o)) = 0.028$, $wR_2(\text{all } F_o^2) = 0.042$; $GooF = 0.934$ for 4048 (1932 unique) reflections and 190 parameters; min./max. residual electron density ± 1.38. Atom parameters are listed in Table 1. Further details of the crystal structure investigation may be obtained from the Fachinformationszentrum Karlsruhe, D-76344 Eggenstein-Leopoldshafen, Germany (e-mail: crysdata@fiz-karlsruhe.de; fax: (+49) 7247-808-666) on quoting the depository number CSD-391323.

Table 1: Atomic coordinates and equivalent displacement parameters U_{eq} of fibrous phosphorus.[a]

Atom	x	y	z	U_{eq} [pm^2]
P(1)	0.1515(5)	0.3308(4)	−0.253(1)	260(10)
P(2)	0.2754(5)	0.4729(5)	−0.2516(9)	260(20)
P(3)	0.1510(5)	0.5801(4)	−0.258(1)	270(10)
P(4)	0.2796(5)	0.7243(4)	−0.240(1)	230(10)
P(5)	0.1603(5)	−0.1640(4)	−0.2606(9)	260(10)
P(6)	0.2876(6)	−0.0284(5)	−0.263(1)	300(20)
P(7)	0.1578(5)	0.0728(4)	−0.2662(9)	270(10)
P(8)	0.2777(6)	0.2260(5)	−0.241(1)	270(20)
P(9)	0.3550(6)	0.4924(4)	0.292(1)	280(20)
P(10)	0.4123(6)	0.5813(5)	0.119(1)	280(20)
P(11)	0.3533(5)	0.7410(4)	0.2912(9)	230(10)
P(12)	0.4152(5)	0.8318(4)	0.1267(9)	240(10)
P(13)	0.3636(5)	−0.0037(4)	0.284(1)	260(10)
P(14)	0.4255(6)	0.0825(4)	0.109(1)	270(10)
P(15)	0.3598(5)	0.2365(5)	0.2835(9)	260(10)
P(16)	0.4144(5)	0.3352(4)	0.1270(9)	220(10)
P(17)	0.1587(6)	0.4055(5)	0.101(1)	340(20)
P(18)	0.1572(5)	0.6844(4)	0.097(1)	310(10)
P(19)	0.1695(5)	−0.0460(4)	0.0915(8)	320(10)
P(20)	0.1655(5)	0.1310(4)	0.0890(8)	310(10)
P(21)	0.0765(3)	0.5517(3)	0.1705(6)	310(10)

[a] Standard deviations for the last significant digit are given in parentheses. U_{eq} (in pm^2) is defined as one-third of the trace of the orthogonalized U_{ij} tensor.

Computational methods: The initial geometries were taken from the X-ray crystal data of Hittorf's[5] and fibrous phosphorus. To investigate the relative stability of these allotropes in the bulk phase, the initial structures were optimized using the density functional based tight-binding method (DFTB)[9] and the Γ-point approximation, as implemented in the program package *deMon*.[20] The experimental lattice constants were kept fixed during the geometry optimizations, and the convergence criterion 10^{-6} atomic units for the energy gradient was used. The validity of the gamma approximation was checked with different k-point sets and different sized supercells. The energy contribution of van der Waals interactions for the allotropes was calculated for fixed (X-ray) and optimized geometries. The optimized bond lengths, bond angles, and dihedral angles are given together with the corresponding experimental values in Tables 2–5 of the Supporting Information. The calculations of the clusters as models for the *[P9]* units were also performed with the program package *deMon*[20] within density functional theory (DFT) method and the local density approximation (LDA) for the exchange-correlation (XC) energy, using a DZVP basis set.

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